

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative researcher specializing in stochastic processes, likelihood-based inference, simulation, and scale- and frequency-resolved signal diagnostics. Builds reproducible Python/PyTorch research systems for large dependent datasets and integrates C++ for performance-critical computation. Brings rigorous hypothesis-testing and model-evaluation experience, with demonstrated interest in systematic trading, market making, and empirical strategy research.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Research Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, SQL, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, cached artifacts, and restartable research pipelines

Quantitative Methods: Probability and stochastic processes, Monte Carlo simulation, hypothesis testing, MLE/GLS, numerical optimization, time-series and spatiotemporal modeling, spectral and autocorrelation analysis, regression, machine learning, model validation, and performance evaluation

QUANTITATIVE INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, translating a monitoring problem into explicit hypotheses, likelihood-ratio boundaries, and controlled sequential decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate boundary-crossing probabilities, stopping behavior, Type I/II error, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Evaluated model performance and feature-attribution patterns and translated findings into pricing and risk-segmentation recommendations for business stakeholders.

QUANTITATIVE RESEARCH AND TRADING

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and evaluated a market-making strategy using market-based fair-value estimation, inventory-aware quoting, rolling-window exposure controls, dynamic liquidation rules, and benchmark-driven performance analysis.

Ph.D. Dissertation Research - Scalable Stochastic Modeling and Signal Diagnostics

Sep 2024 - Present

- **Scalable Inference:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, using ordered conditional models to fit up to 145,008 observations per day.
- **Signal Diagnostics:** Designed scale- and frequency-resolved analyses to determine whether apparent patterns reflect persistent dependence, high-frequency noise, missingness, acquisition artifacts, or model misspecification.
- **Research Infrastructure:** Built reusable Python/PyTorch workflows for data-quality filtering, time-dependent alignment, optimization, simulation, cached outputs, and restartable CPU/GPU and HPC execution; integrated performance-critical C++ ordering routines through pybind11.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)